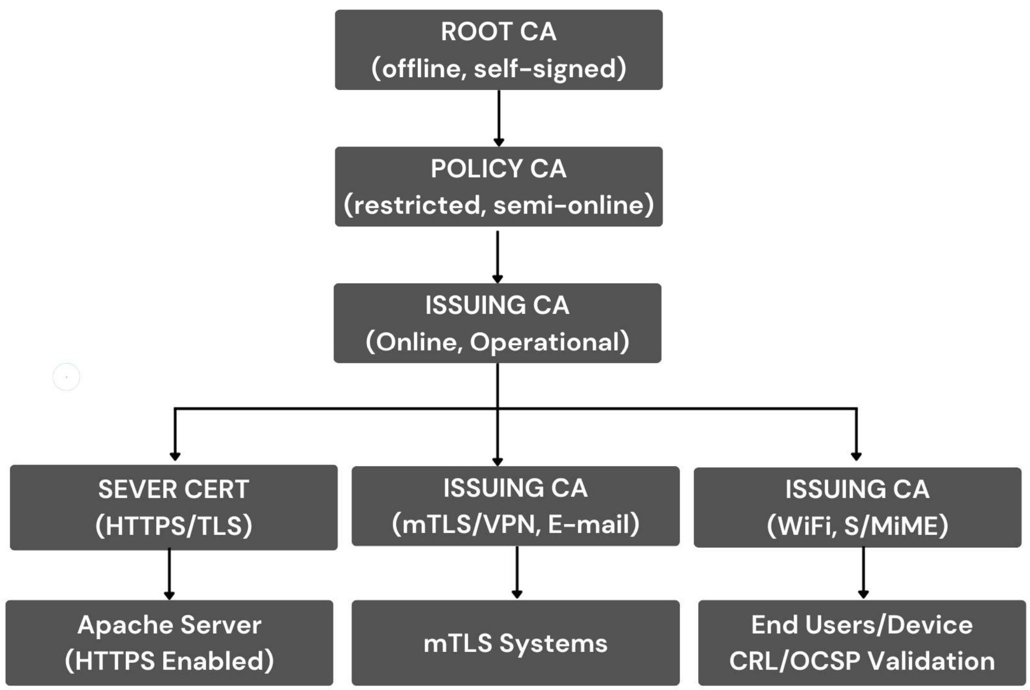
Public Key Infrastructure (PKI)

Public Key Infrastructure (PKI) is a trust-based system that combines cryptography, policies, processes, hardware, software, and human roles to securely manage digital certificates and public–private key pairs. The main goal of PKI is to make sure that digital communication is secure and trustworthy. Using PKI, systems can verify identities, encrypt data, protect integrity, and ensure non-repudiation. In practical environments, PKI is used in HTTPS websites, VPNs, secure email, WiFi authentication, mutual TLS (mTLS), enterprise networks, and government infrastructures.

Here, I have implemented 3-Tier PKI architecture. This model is preferred in real-world deployments because it separates trust, policy, and operational responsibilities. This separation increases security and reduces the impact of compromise.



Root Certificate Authority (Root CA)

The Root CA is the highest authority in the PKI hierarchy. It is the foundation of trust for the entire system. The Root CA certificate is self-signed, which means it does not depend on any other authority for trust. All other certificates in the PKI ultimately trace their trust back to the Root CA.

Key responsibilities of the Root CA:

- Acts as the trust anchor for the PKI
- Signs only Intermediate / Policy CA certificates
- Does not issue certificates to end users or servers

The Root CA is extremely sensitive. Because compromise of the Root CA breaks the entire PKI, it is kept offline and used very rarely. Its private key is usually stored in an HSM or an isolated offline system. Root CA certificates typically have a long validity period (10–20 years).

Intermediate / Policy Certificate Authority

The Intermediate CA sits between the Root CA and the Issuing CA. It is signed by the Root CA and acts as a policy enforcement and protection layer. This CA helps protect the Root CA from exposure while allowing the PKI to scale.

Role of the Intermediate CA:

- Enforces certificate policies and constraints
- Signs Issuing CA certificates
- Separates trust from daily operations

If an Issuing CA is compromised, the Root CA remains safe. Even if an Intermediate CA is compromised, the damage is limited to that branch only. This is one of the main reasons why enterprise and government PKIs always use an Intermediate layer. The Intermediate CA usually has a medium validity period (3–5 years) and may be kept online only when required.

Issuing Certificate Authority (Issuing CA)

The Issuing CA is the operational component of the PKI. It is signed by the Intermediate CA and is responsible for issuing certificates to end entities such as servers, clients, users, and devices.

Responsibilities of the Issuing CA:

- Issues server, client, user, and device certificates
- Publishes Certificate Revocation Lists (CRL)
- Supports Online Certificate Status Protocol (OCSP)

The Issuing CA is the most exposed part of the PKI because it interacts directly with users and systems. For this reason, it has a shorter validity period and is carefully monitored.

Server Certificates (HTTPS / TLS)

Server certificates are used to identify servers and enable encrypted communication using TLS. These certificates are issued by the Issuing CA and contain domain names or IP addresses in the Subject Alternative Name (SAN) field.

How it works in practical:

- A web server generates a key pair and CSR
- The Issuing CA signs and issues the server certificate
- The server presents the certificate during the TLS handshake

Example: When a user visits an HTTPS website, the browser verifies the certificate chain (Issuing → Intermediate → Root). If the chain is trusted and valid, an encrypted TLS session is established.

Client Certificates (mTLS – Client Identity)

Client certificates are used to identify clients rather than servers. They are a key component of mutual TLS (mTLS), where both the client and the server authenticate each other.

Common use cases:

- mTLS for service-to-service authentication
- VPN client authentication
- Zero Trust environments

Client certificates provide stronger security than passwords because identity is tied to cryptographic keys.

User and Device Certificates

User and device certificates provide digital identities to people and machines. These certificates are also issued by the Issuing CA.

Typical use cases:

- WiFi authentication (802.1X)
- Secure email using S/MIME
- Corporate user authentication
- IoT device identity

These certificates help eliminate password-based authentication and improve overall security.

Certificate Revocation (CRL & OCSP)

Certificates may need to be revoked before their expiration if a private key is compromised, a device is lost, or a user leaves the organization.

Certificate Revocation List (CRL):

- A list of revoked certificate serial numbers
- Published periodically by the Issuing CA
- Clients download and check the list

Online Certificate Status Protocol (OCSP):

- Provides real-time certificate status checking
- Client queries an OCSP responder
- Faster and more efficient than CRL

Revocation ensures that untrusted or compromised certificates are rejected even if they have not expired.

How PKI Works End-to-End (Flow)

The PKI workflow starts by establishing trust. The Root CA certificate is installed in operating systems, browsers, or application trust stores. The Root CA signs the Intermediate CA, and the Intermediate CA signs the Issuing CA. When a server, client, or user needs a certificate, it generates a key pair and submits a Certificate Signing Request (CSR) to the Issuing CA. After identity verification, the Issuing CA issues the certificate. During communication, the receiving system verifies the certificate chain, checks validity, confirms key usage, and performs revocation checks using CRL or OCSP. If all checks succeed, secure communication is established. If any check fails, the connection is rejected. In the given below,

Implementation Walkthrough

Step 1: Environment Preparation
#sudo apt update
#sudo apt install openssl -y
#sudo apt install apache2 -y

PKI operations require cryptographic tools and a real service to deploy certificates. OpenSSL provides PKI functionalities, and Apache is used to demonstrate TLS in a real-world web server scenario. **apt update** refreshes package indexes. **apt install** downloads and installs OpenSSL and Apache along with their dependencies from trusted repositories.

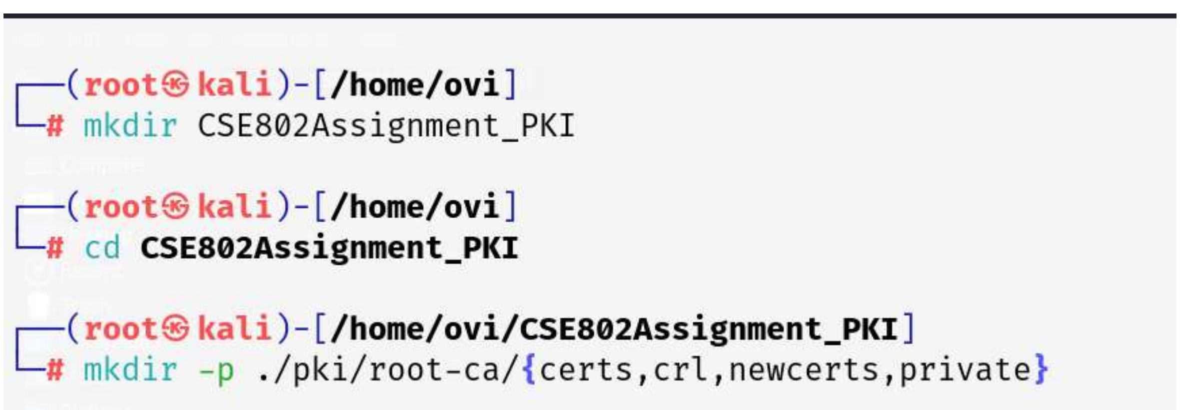
#openssl version
#apache2 -v

It verifies that the required software is correctly installed before starting PKI operations. Each command queries the binary and prints the installed version directly from the system.

Step 2: PKI Directory Structure

i. Root-ca

First, we create a dedicated working directory for this task and name it CSE802Assignment_PKI. After creating it, we move into this directory so that all PKI-related files remain organized in one place. Inside the main working directory, we then create a directory for the Root CA. Within the *Root-ca* directory, several sub-directories are created such as certs, crl, newcerts, and private, along with essential files like index.txt and serial. These components are required for certificate management, tracking issued certificates, and maintaining the CA database.
The same directory structure is then repeated for the Policy CA, Issuing CA, and the Server directory.



```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# touch ./pki/root-ca/index.txt

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# echo 1000 > ./pki/root-ca/serial
```

After setting up the directory structure, we generate the Root.key file, which is the private key of the Root Certificate Authority (Root CA). This key is created using the AES-256 encryption algorithm to ensure strong protection of the private key. The Root private key is used to self-sign the Root CA certificate and to sign the certificates of Intermediate (Policy) CAs. Since the Root CA is the trust anchor of the entire PKI, this key is kept highly secure and is usually stored offline.

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl genrsa -aes256 -out ./pki/root-ca/private/root.key 4096
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
```

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl req -x509 -new -key ./pki/root-ca/private/root.key \
  -sha256 -days 7300 \
  -out ./pki/root-ca/certs/root-ca.crt

Enter pass phrase for ./pki/root-ca/private/root.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:BD
State or Province Name (full name) [Some-State]:Dhaka
Locality Name (eg, city) []:Dhaka
Organization Name (eg, company) [Internet Widgits Pty Ltd]:CSEDU
Organizational Unit Name (eg, section) []:PMICS
Common Name (e.g. server FQDN or YOUR name) []:CSEDU Root Certification Authority
Email Address []:ovishek.54@student.cse.du.ac.bd
```

Generates a self-signed root certificate valid for ~20 years.

DN details: Country: BD, State: Dhaka, Locality: Dhaka, Organization: CSEDU, Organizational Unit: PMICS, Common Name: CSEDU Root Certification Authority, Email: ovishek.54@student.cse.du.ac.bd

We provide the required certificate information to identify the Root CA, after which OpenSSL generates a self-signed Root CA certificate. Finally, the generated root-ca.crt file is checked to confirm successful certificate creation.

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl x509 -in ./pki/root-ca/certs/root-ca.crt -noout -text

Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            1b:44:02:f7:0b:07:a1:7c:0c:2d:2b:57:0f:9f:6e:3b:00:2c:7f:5f
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Root Certification Authority, emailAddress=ovishek.
54@student.cse.du.ac.bd
        Validity
            Not Before: Dec 15 02:26:36 2025 GMT
            Not After : Dec 10 02:26:36 2045 GMT
        Subject: C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Root Certification Authority, emailAddress=ovishek
.54@student.cse.du.ac.bd
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (4096 bit)
                Modulus:
                    00:b8:de:47:e8:83:9e:57:ca:fe:f7:30:63:47:ce:
                    e9:97:41:7d:0e:49:5e:18:1a:40:ed:75:e9:fc:04:
                    1d:a6:5b:5f:fd:6c:ee:5d:b8:ee:f7:ab:c7:ae:59:
                    f1:3e:d2:f7:16:06:9c:9f:aa:de:ab:c0:2d:f1:36:
                    11:3e:9c:f5:72:43:29:73:b4:ba:41:e9:da:02:b8:
                    8b:19:b9:8d:f3:81:19:09:3d:83:54:75:19:c5:09:
                    25:4e:65:3d:e6:73:93:08:ab:67:37:c3:4a:0d:15:
                    8b:f1:a1:08:3e:06:e0:94:a3:3d:ee:c1:bf:67:4c:
                    94:2a:42:3a:44:40:2e:46:93:d9:25:d6:05:07:2e:
                    36:07:a6:58:ba:7a:20:60:12:4f:ea:1f:82:af:93:
                    84:62:9a:57:c1:35:fe:81:df:9e:73:28:8a:0e:7d:
                    04:ab:b5:bd:62:20:f8:69:49:02:9d:5a:ca:92:dc:
                    63:b8:ce:cc:e5:78:4c:1b:38:2d:6f:66:79:57:f7:
                    10:fd:a9:7e:84:0f:68:c7:27:e8:a0:de:10:d5:1d:
                    f1:50:99:1a:fc:8f:58:ad:19:e6:69:5f:5f:58:4e:
```

ii. Policy-ca

After creating the Policy CA directory structure, a secure private key (policy.key) is generated using strong encryption. This key is used to generate the Policy CA certificate and to sign Issuing CA certificates.

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# mkdir -p ./pki/policy-ca/{certs,crl,newcerts,private}

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# touch ./pki/policy-ca/index.txt

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# echo 2000 > ./pki/policy-ca/serial

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# echo 1000 > ./pki/policy-ca/crlnumber

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl genrsa -aes256 -out ./pki/policy-ca/private/policy.key 4096
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
```

Based on the generated encrypted private key (policy.key), a certificate for the Policy CA is created. This certificate is signed by the Root CA, allowing the Policy CA to inherit trust from the Root CA and become part of the PKI trust chain.

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl req -new -sha256 \
    -key ./pki/policy-ca/private/policy.key \
    -out ./pki/policy-ca/policy.csr

Enter pass phrase for ./pki/policy-ca/private/policy.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:BD
State or Province Name (full name) [Some-State]:Dhaka
Locality Name (eg, city) []:Dhaka
Organization Name (eg, company) [Internet Widgits Pty Ltd]:CSEDU
Organizational Unit Name (eg, section) []:PMICS
Common Name (e.g. server FQDN or YOUR name) []:CSEDU Policy Certification Authority
Email Address []:ovishek.54@student.cse.du.ac.bd

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:123456
An optional company name []:PMICS5
```

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl ca -config ./pki/root-ca/openssl.cnf \
    -extensions v3_intermediate_ca \
    -days 3650 -notext -md sha256 \
    -in ./pki/policy-ca/policy.csr \
    -out ./pki/policy-ca/certs/policy.crt

Using configuration from ./pki/root-ca/openssl.cnf
Enter pass phrase for /home/ovi/CSE802Assignment_PKI/pki/root-ca/priv
ate/root.key:
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
countryName           :PRINTABLE:'BD'
stateOrProvinceName   :ASN.1 12:'Dhaka'
localityName          :ASN.1 12:'Dhaka'
organizationName      :ASN.1 12:'CSEDU'
organizationalUnitName:ASN.1 12:'PMICS'
commonName            :ASN.1 12:'CSEDU Policy Certification Authority
'
Certificate is to be certified until Dec 13 02:58:14 2035 GMT (3650 d
ays)
Sign the certificate? [y/n]:y


1 out of 1 certificate requests certified, commit? [y/n]y
Write out database with 1 new entries
Database updated
```

iii. Issuing-ca

After creating the Issuing CA directory structure, the private key for the Issuing CA, named issuing.key, is generated using strong cryptographic protection. Using this private key, a certificate request for the Issuing CA is created. The Policy CA then signs the Issuing CA certificate, allowing it to inherit trust from the upper layers of the PKI hierarchy. Finally, the Issuing CA certificate is verified to ensure that it has been correctly issued and linked to the Policy and Root CAs.

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# mkdir -p ./pki/issuing-ca/{certs,crl,newcerts,private}
```

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl genrsa -aes256 \
-out ./pki/issuing-ca/private/issuing.key 4096

Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
```

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl req -new -sha256 \
-key ./pki/issuing-ca/private/issuing.key \
-out ./pki/issuing-ca/issuing.csr

Enter pass phrase for ./pki/issuing-ca/private/issuing.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:BD
State or Province Name (full name) [Some-State]:Dhaka
Locality Name (eg, city) []:Dhaka
Organization Name (eg, company) [Internet Widgits Pty Ltd]:CSEDU
Organizational Unit Name (eg, section) []:PMICS
Common Name (e.g. server FQDN or YOUR name) []:CSEDU Issue Certificate Authority
Email Address []:ovishek.54@student.cse.du.ac.bd

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:123456
An optional company name []:PMICS5
```

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl x509 -in ./pki/issuing-ca/certs/issuing.crt -noout -text

Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number: 8192 (0x2000)
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=BD, ST=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Policy Certification Authority
        Validity
            Not Before: Dec 15 03:28:21 2025 GMT
            Not After : Dec 13 03:28:21 2035 GMT
        Subject: C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Issue Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac
.bd
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (4096 bit)
                Modulus:
                    00:b8:97:1b:52:82:b3:9c:d9:cb:40:a5:21:54:55:
                    1a:47:7b:fe:72:54:b9:e2:f9:77:76:af:b2:14:71:
                    fd:0b:2c:8e:3f:fd:f2:71:8f:a4:0a:bc:6d:2b:b7:
                    b7:5d:f1:a9:3b:65:54:82:34:0a:0b:e0:91:03:1a:
                    cd:71:bd:78:0f:23:cb:54:7e:90:04:3f:05:52:cb:
                    43:50:30:19:24:35:89:cc:60:b9:69:75:e5:e0:c2:
                    6e:fb:99:c7:4d:90:e1:47:77:ee:3a:77:b0:35:25:
                    e9:0c:53:04:08:08:3d:29:29:77:b0:71:7b:0d:be:
                    49:de:43:60:3f:ab:3b:65:23:07:e4:80:98:5f:95:
```

Step 3: Sign Server Certificate with Issuing CA (Apache Web Server)

i. Directory Setup for Server Certificate

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# mkdir -p ./pki/server/{certs,crl,newcerts,private}

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# touch ./pki/server/index.txt

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# echo 1000 > ./pki/server/serial

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# echo 1000 > ./pki/server/crlnumber
```

ii. Generate Server Key

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl genrsa -out ./pki/server/private/server.key 4096
```

iii. CSR for Server Certificate

CSR DN:

- CN: CSEDU Server Certificate Authority
- Email: ovishek.54@student.cse.du.ac.bd
- Challenge password: 123456
- Optional company: PMICSS

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl req -new -sha256 \
-key ./pki/server/private/server.key \
-out ./pki/server/server.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:BD
State or Province Name (full name) [Some-State]:Dhaka
Locality Name (eg, city) []:Dhaka
Organization Name (eg, company) [Internet Widgits Pty Ltd]:CSEDU
Organizational Unit Name (eg, section) []:PMICS
Common Name (e.g. server FQDN or YOUR name) []:CSEDU Server Certificate Authority
Email Address []:ovishek.54@student.cse.du.ac.bd

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:123456
An optional company name []:PMICS5
```

iv. View Server CSR Details: Shows CSR subject and public key details.

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl req -in ./pki/server/server.csr -noout -text

Certificate Request:
    Data:
        Version: 1 (0x0)
        Subject: C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Server Certificate Authority, emailAddress=ovishek.54@student.cse.du.a
c.bd
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (4096 bit)
                Modulus:
                    00:9c:b7:3a:1b:fe:1f:e7:b1:d6:53:a2:43:fa:85:
                    01:44:8f:1a:a2:c9:01:1b:15:42:be:38:22:94:e1:
                    63:77:d2:94:89:d0:6f:3b:30:f6:64:85:59:39:4e:
                    f2:30:f0:fd:df:95:2a:d8:51:f3:3b:48:47:90:84:
                    e5:af:85:46:8e:e1:db:35:3e:61:37:e4:ce:3c:b6:
                    1a:f6:37:52:93:5e:b5:92:18:25:ce:9a:d0:a2:77:
                    cc:a9:00:c4:f8:9b:8d:67:c3:00:2f:b0:ef:b3:8d:
                    5c:86:50:6d:64:8d:0a:ad:66:46:90:05:b4:c8:f1:
                    e4:b1:d1:71:b3:e6:48:5f:be:92:af:a6:df:85:29:
                    00:6c:14:28:c5:31:c9:81:31:50:5e:36:aa:aa:54:
                    e7:8d:0d:f5:ae:b6:5b:44:40:bd:07:84:84:3e:a6:
                    64:41:ea:87:1d:49:2c:55:04:f3:df:fd:bf:57:db:
                    7e:6a:1c:d8:eb:d3:f9:0f:e6:c6:85:94:7e:ad:b8:
                    89:4c:f8:78:a4:5d:1e:fc:df:7c:7f:00:dc:8f:36:
                    79:fc:53:55:1b:15:a8:ec:88:07:88:e9:da:92:9b:
                    aa:63:f9:a0:9e:ca:e3:5e:75:08:dc:73:12:48:b1:
                    26:49:f7:34:8a:12:da:eb:11:96:c4:88:59:06:8c:
```

v. Sign Server Certificate with Issuing CA

- Issuing CA signs the server certificate.
- Valid for ~2.26 years (825 days).

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl ca -config ./pki/issuing-ca/openssl.cnf \
-extensions server_cert \
-days 825 -notext -md sha256 \
-in ./pki/server/server.csr \
-out ./pki/server/server.crt

Using configuration from ./pki/issuing-ca/openssl.cnf
Enter pass phrase for /home/ovi/CSE802Assignment_PKI/pki/issuing-ca/private/issuing.key:
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
countryName           :PRINTABLE:'BD'
stateOrProvinceName   :ASN.1 12:'Dhaka'
localityName          :ASN.1 12:'Dhaka'
organizationName      :ASN.1 12:'CSEDU'
organizationalUnitName:ASN.1 12:'PMICS'
commonName            :ASN.1 12:'CSEDU Server Certificate Authority'
emailAddress          :IA5STRING:'ovishek.54@student.cse.du.ac.bd'
Certificate is to be certified until Mar 19 03:55:27 2028 GMT (825 days)
Sign the certificate? [y/n]:y


1 out of 1 certificate requests certified, commit? [y/n]y
Write out database with 1 new entries
Database updated
```

vi. View Server Certificate Details: Displays server certificate details.

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl x509 -in ./pki/server/server.crt -noout -text

Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number: 12288 (0x3000)
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Issue Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.
bd
        Validity
            Not Before: Dec 15 03:55:27 2025 GMT
            Not After : Mar 19 03:55:27 2028 GMT
        Subject: C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Server Certificate Authority, emailAddress=ovishek.54@student.cse.du.a
c.bd
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (4096 bit)
                Modulus:
                    00:9c:b7:3a:1b:fe:1f:e7:b1:d6:53:a2:43:fa:85:
                    01:44:8f:1a:a2:c9:01:1b:15:42:be:38:22:94:e1:
                    63:77:d2:94:89:d0:6f:3b:30:f6:64:85:59:39:4e:
                    f2:30:f0:fd:df:95:2a:d8:51:f3:3b:48:47:90:84:
                    e5:af:85:46:8e:e1:db:35:3e:61:37:e4:ce:3c:b6:
                    1a:f6:37:52:93:5e:b5:92:18:25:ce:9a:d0:a2:77:
                    cc:a9:00:c4:f8:9b:8d:67:c3:00:2f:b0:ef:b3:8d:
                    5c:86:50:6d:64:8d:0a:ad:66:46:90:05:b4:c8:f1:
                    e4:b1:d1:71:b3:e6:48:5f:be:92:af:a6:df:85:29:
```

Step 4: Create Full Certificate Chain

It combines certificates into a single chain file for server use.

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# cat ./pki/server/server.crt \
  ./pki/issuing-ca/certs/issuing.crt \
  ./pki/policy-ca/certs/policy.crt \
  ./pki/root-ca/certs/root-ca.crt \
> ./pki/server/server-full-chain.crt
```

Step 5: Restart and Check the Dependencies

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# apache2 -v

Server version: Apache/2.4.65 (Debian)
Server built:   2025-08-15T08:30:58

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Module socache_shmcb already enabled
Module ssl already enabled

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# systemctl restart apache2

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# apache2ctl -M | grep ssl

 ssl_module (shared)
```

Step 6: Configure the Default SSL configuration file and update the directory with our project directory

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# nano /etc/apache2/sites-available/default-ssl.conf

┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# nano /etc/apache2/sites-available/default-ssl.conf
```

```
 GNU nano 8.6                    /etc/apache2/sites-available/default-ssl.conf
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf

        #   SSL Engine Switch:
        #   Enable/Disable SSL for this virtual host.
        SSLEngine on

        #   A self-signed (snakeoil) certificate can be created by installing
        #   the ssl-cert package. See
        #   /usr/share/doc/apache2/README.Debian.gz for more info.
        #   If both key and certificate are stored in the same file, only the
        #   SSLCertificateFile directive is needed.
        SSLCertificateFile      /home/ovi/CSE802Assignment_PKI/apache_ssl/server-full-chain.crt
        SSLCertificateKeyFile   /home/ovi/CSE802Assignment_PKI/apache_ssl/server.key

        #   Server Certificate Chain:
        #   Point SSLCertificateChainFile at a file containing the
        #   concatenation of PEM encoded CA certificates which form the
        #   certificate chain for the server certificate. Alternatively
        #   the referenced file can be the same as SSLCertificateFile
        #   when the CA certificates are directly appended to the server
        #   certificate for convinience.
        #SSLCertificateChainFile /etc/apache2/ssl.crt/server-ca.crt
```

Step 7: Finally Verify the certificates in HTTPS Server in the Terminal and Browser

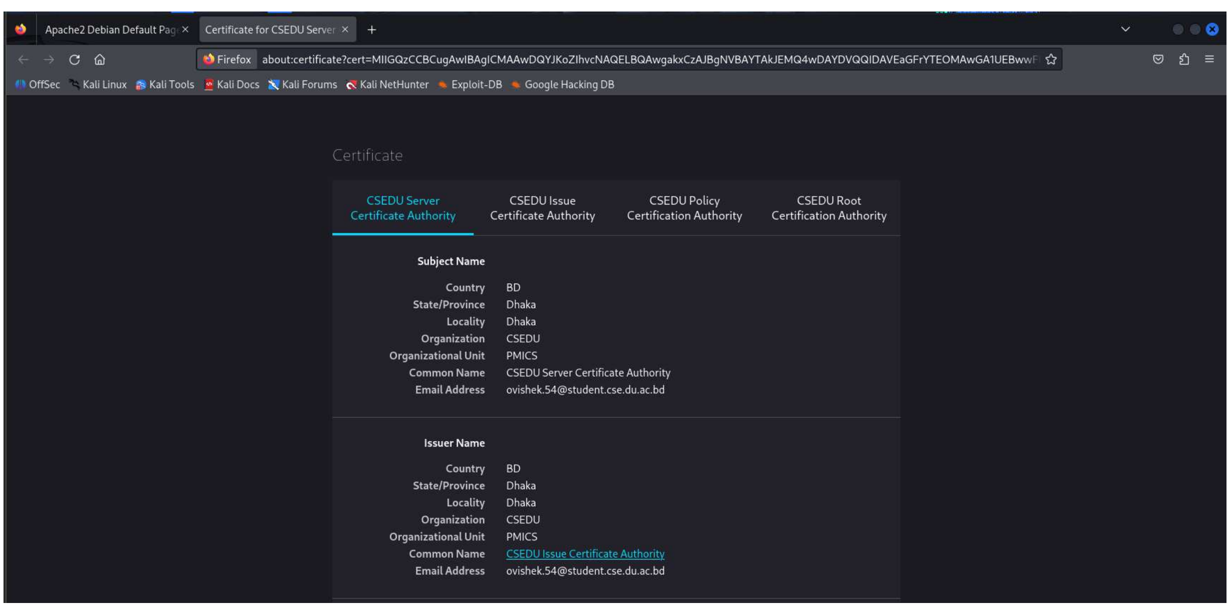
```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl s_client -connect localhost:443 -showcerts

Connecting to 127.0.0.1
CONNECTED(00000003)
Can't use SSL_get_servername
depth=3 C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Root Certification Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
verify error:num=19:self-signed certificate in certificate chain
verify return:1
depth=3 C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Root Certification Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
verify return:1
depth=2 C=BD, ST=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Policy Certification Authority
verify return:1
depth=1 C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Issue Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
verify return:1
depth=0 C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Server Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
verify return:1
---
Certificate chain
 0 s:C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Server Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
   i:C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Issue Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
   a:PKEY: RSA, 4096 (bit); sigalg: sha256WithRSAEncryption
   v:NotBefore: Dec 15 03:55:27 2025 GMT; NotAfter: Mar 19 03:55:27 2028 GMT
```

```
izjVQcxShftvV1L8GRz1S0uSc1LXO67vEgnAgI1NTG0W1SIorFJH2FpoOnIoo05/
dEM+0Kk/6SpDTdPUy/Niv9t5jcmxf+GNH1iSxNpmERe4ZuZja7FXH9xQ0DTKLhF4
FUDdfqUv+zI6Ks3bHVlvQuBawUzVoeuW5U19sz5bCTYztnPsPQWWgYutLdl9w2Os
/mughNnN7RJMHTEM0ShLdgK0/5FXpBzsiyJj7gCPaH3UGE32GsxPhOGlEOOpaV6v
tvzuabz2JfnQUMpmLMrudnM9A7cEpMliYODSrhGdkIOomM8oUxNBtE0+ZI67gKLI
Bhv60WiUKINvPhQT7cRJcg37x/UHvbA=
-----END CERTIFICATE-----
 1 s:C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Issue Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
   i:C=BD, ST=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Policy Certification Authority
   a:PKEY: RSA, 4096 (bit); sigalg: sha256WithRSAEncryption
   v:NotBefore: Dec 15 03:28:21 2025 GMT; NotAfter: Dec 13 03:28:21 2035 GMT
-----BEGIN CERTIFICATE-----
MIIF+DCCA+CgAwIBAgICIAAwDQYJKoZIhvcNAQELBQAwbDELMAkGA1UEBhMCQkQx
DjAMBgNVBAgMBURoYWthMQ4wDAYDVQQKDAVDU0VEVTEOMAwGA1UECwwFUE1JQ1Mx
LTArBgNVBAMMJENTRURVIFBvbGljeSBDZXJ0aWZpY2F0aW9uIEF1dGhvcml0eTAe
Fw0yNTEyMTUwMzI4MjFaFw0zNTEyMTMwMzI4MjFaMIGpMQswCQYDVQQGEwJCRDEO
MAwGA1UECAwFRGhha2ExDjAMBgNVBAcMBURoYWthMQ4wDAYDVQQKDAVDU0VEVTEO
MAwGA1UECwwFUE1JQ1MxKjAoBgNVBAMMIUNTRURVIElzc3VlIENlcnRpZmljYXRl
```

```
1EGMRfGZCKXZuVU1K3/Xd+8GWALEnZNP1KltZ8sY/8WVPUOKsjqgEdmDQEl/HWrz
MBY4U8GfN1rqIYXwNHJY1TK3FEAfow0wKt9xmMjvjq68k/1cDAq7Ei93zFg=
-----END CERTIFICATE-----
 2 s:C=BD, ST=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Policy Certification Authority
   i:C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Root Certification Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
   a:PKEY: RSA, 4096 (bit); sigalg: sha256WithRSAEncryption
   v:NotBefore: Dec 15 02:58:14 2025 GMT; NotAfter: Dec 13 02:58:14 2035 GMT
-----BEGIN CERTIFICATE-----
MIIF+TCCA+GgAwIBAgICEAAwDQYJKoZIhvcNAQELBQAwgaoxCzAJBgNVBAYTAkJE
MQ4wDAYDVQQIDAVEaGFrYTEOMAwGA1UEBwwFRGhha2ExDjAMBgNVBAoMBUNTRURV
MQ4wDAYDVQQLDAVQTUlDUzErMCkGA1UEAwwiQ1NFRFUgUm9vdCBDZXJ0aWZpY2F0
```

```
X3PnxxWZkNg62G1we8F6LKXAOebWwW08yiHE+2oUU87+8jG/MLW5NFWrT77r4gNd
DtXcLrLHJQ9QHdZQR8W8yaC08872MHrdtVE8RZ4N1b6pKY81qY7UuLretowW
-----END CERTIFICATE-----
 3 s:C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Root Certification Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
   i:C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Root Certification Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
   a:PKEY: RSA, 4096 (bit); sigalg: sha256WithRSAEncryption
   v:NotBefore: Dec 15 02:26:36 2025 GMT; NotAfter: Dec 10 02:26:36 2045 GMT
-----BEGIN CERTIFICATE-----
MIIGNzCCBB+gAwIBAgIUG0QC9wsHoXwMLStXD59uOwAsf18wDQYJKoZIhvcNAQEL
BQAwgaoxCzAJBgNVBAYTAkJEMQ4wDAYDVQQIDAVEaGFrYTEOMAwGA1UEBwwFRGhh
a2ExDjAMBgNVBAoMBUNTRURVMQ4wDAYDVQQLDAVQTUlDUzErMCkGA1UEAwwiQ1NF
```

```
-----END CERTIFICATE-----
---
Server certificate
subject=C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Server Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
issuer=C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Issue Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
---
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: rsa_pss_rsae_sha256
Negotiated TLS1.3 group: X25519MLKEM768
---
SSL handshake has read 8182 bytes and written 1740 bytes
Verification error: self-signed certificate in certificate chain
---
```



Revoke Server Certificate

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl ca -config ./pki/issuing-ca/openssl.cnf \
-revoke ./pki/server/server.crt

Using configuration from ./pki/issuing-ca/openssl.cnf
Enter pass phrase for /home/ovi/CSE802Assignment_PKI/pki/issuing-ca/private/issuing.key:
Revoking Certificate 3000.
Database updated
```

Generate CRL for Issuing CA

- Output path seems misnamed (should be .crl).
- Generates a Certificate Revocation List.

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl ca -config ./pki/issuing-ca/openssl.cnf \
-gencrl -out ./pki/issuing-ca/crl/issuing-ca.crl

Using configuration from ./pki/issuing-ca/openssl.cnf
Enter pass phrase for /home/ovi/CSE802Assignment_PKI/pki/issuing-ca/private/issuing.key:
```

View CRL Details

- Displays revoked certificates and CRL details.

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl crl -in ./pki/issuing-ca/crl/issuing-ca.crl -noout -text

Certificate Revocation List (CRL):
        Version 2 (0×1)
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Issue Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.
bd
        Last Update: Dec 15 04:33:53 2025 GMT
        Next Update: Jan 14 04:33:53 2026 GMT
        CRL extensions:
            X509v3 Authority Key Identifier:
                9D:67:A4:F2:D3:F6:E2:AD:05:C2:A1:4E:54:C5:9B:CB:08:9B:26:1E
            X509v3 CRL Number:
                12288
Revoked Certificates:
    Serial Number: 3000
        Revocation Date: Dec 15 04:24:56 2025 GMT
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
        82:92:de:ad:b1:93:f5:7d:4c:ee:5d:d0:fd:58:99:f7:2d:1b:
        11:3b:e3:ad:3d:43:0d:d7:2a:85:dc:f2:2f:d9:09:83:fe:b3:
        ee:de:2c:db:8a:cf:e2:56:62:0a:1d:a5:bd:6a:58:e4:ef:69:
```

Verify Certificate with CRL & CRT Check


```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl crl -in ./pki/issuing-ca/crl/issuing-ca.crl -noout -text

Certificate Revocation List (CRL):
        Version 2 (0×1)
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Issue Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.
bd
        Last Update: Dec 15 04:33:53 2025 GMT
        Next Update: Jan 14 04:33:53 2026 GMT
        CRL extensions:
            X509v3 Authority Key Identifier:
                9D:67:A4:F2:D3:F6:E2:AD:05:C2:A1:4E:54:C5:9B:CB:08:9B:26:1E
            X509v3 CRL Number:
                12288
Revoked Certificates:
    Serial Number: 3000
        Revocation Date: Dec 15 04:24:56 2025 GMT
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
        82:92:de:ad:b1:93:f5:7d:4c:ee:5d:d0:fd:58:99:f7:2d:1b:
        11:3b:e3:ad:3d:43:0d:d7:2a:85:dc:f2:2f:d9:09:83:fe:b3:
        ee:de:2c:db:8a:cf:e2:56:62:0a:1d:a5:bd:6a:58:e4:ef:69:
```

```
┌──(root㉿kali)-[/home/ovi/CSE802Assignment_PKI]
└─# openssl verify -crl_check -CAfile ./pki/ca-full-chain.crt \
-CRLfile ./pki/issuing-ca/crl/issuing-ca.crl ./pki/server/server.crt

C=BD, ST=Dhaka, L=Dhaka, O=CSEDU, OU=PMICS, CN=CSEDU Server Certificate Authority, emailAddress=ovishek.54@student.cse.du.ac.bd
error 23 at 0 depth lookup: certificate revoked
```

A 3-Tier PKI provides a secure and scalable trust model by isolating the Root CA, enforcing policies through the Intermediate CA, and handling daily certificate operations with the Issuing CA. This layered approach minimizes risk while supporting secure services such as HTTPS, mTLS, VPNs, WiFi authentication, and secure email. This is why 3-Tier PKI is the standard architecture used in enterprise and government environments.